

Day 3 – VirtualBox Networking

Sabse Pehle Basic Concept Clear Karein

Network Kya Hota Hai?

Jab 2 ya zyada devices ek dusre se connect hote hain → usko network kehte hain.

Example:

- Tumhara mobile 
- Tumhara laptop 
- Wi-Fi router 

Ye sab ek network me hain.

Host vs VM

Host

Tumhara original computer (Windows/Linux).

VM (Virtual Machine)

Virtual computer jo VirtualBox ke andar chal raha hai.

Socho:

Host = Ghar 

VM = Ghar ke andar ek alag room 

Ab question:

Room ko internet milega ya nahi?

Room dusre room se baat karega ya nahi?

 Ye decide karta hai **Network Mode**.

1. NAT – Sabse Simple Mode

Real Life Example

Ghar me:

- 5 mobile

- 2 laptops
- 1 TV

Sab ek hi Wi-Fi router se internet use karte hain.

Internet ko lagta hai:

👉 Sirf 1 IP (router ka IP)

Bilkul waise hi NAT kaam karta hai.

NAT Me Kya Hota Hai?

VM internet use karta hai
par internet ko pata nahi hota ki VM exist karta hai.

Internet ko lagta hai:

👉 Sab traffic host se aa raha hai.

VM chhup jata hai host ke peeche.

Visual Samjho

Internet 



Host  (Real IP)



VM  (Hidden)

NAT Me Allowed

- ✓ VM → Internet = YES
- ✗ Internet → VM = NO
- ✗ Host → VM = NO
- ✗ VM → Host = NO
- ✗ VM ↔ VM = NO

Kab Use Kare?

Jab sirf:

- Linux update karna ho
- Tools download karna ho
- Browsing karni ho

👉 Bas internet chahiye → NAT



2 NAT Methods - NAT Ke Smart Version



2. NAT NETWORK – NAT Ka Smart version

NAT me problem kya thi?

👉 VM dusre VM se baat nahi kar sakta tha.

Isliye aaya NAT Network.



Simple Example

Socho ek building hai:

- Room 1 (VM1)
- Room 2 (VM2)
- Room 3 (VM3)

Sabko internet bhi chahiye
aur ek dusre se baat bhi karni hai.

👉 NAT Network ye allow karta hai.



Visual Samjho

Internet 🌐



Host (NAT Gateway) 🖥️



VM1 🖥️ ↔ VM2 🖥️ ↔ VM3 🖥️



NAT Network Me Allowed

✓ VM → Internet = YES

✓ VM ↔ VM = YES

✗ Internet → VM = NO

✗ Host → VM = NO



Kab Use Kare?

Cybersecurity lab me:

- Kali Linux
- Metasploitable
- Web Server VM

Sab ek dusre se connect hone chahiye
aur internet bhi mile.

👉 Use NAT Network

3. Bridged Adapter – Real Computer Mode

Ye sabse powerful mode hai.

Isme VM:

👉 Real laptop ki tarah behave karta hai.

Router usko alag IP deta hai.

Real Life Example

Ghar me:

- Laptop = 192.168.1.5
- Mobile = 192.168.1.8
- TV = 192.168.1.12

Bridged me:

VM bhi ban jayega:

👉 192.168.1.20 (example)

Visual

Router 📶



Host 🖥️ (192.168.1.10)

VM 🖥️ (192.168.1.20)

Iska Matlab

Network ke sab log:

- ✓ VM ko dekh sakte hain
 - ✓ Ping kar sakte hain
 - ✓ SSH kar sakte hain
 - ✓ Web server open kar sakte hain
-

Bridged Me Allowed

- ✓ VM → Internet
- ✓ Host → VM
- ✓ VM → Host
- ✓ LAN → VM
- ✓ VM ↔ VM

Sab allowed.

Danger

Public Wi-Fi pe use kiya?

Agar firewall nahi hai?

👉 VM hack ho sakta hai.

Ye mode secure lab ke liye nahi hai.

Kab Use Kare?

- Web server test karna
 - Real LAN testing
 - Networking practice
-

4. Host-Only Adapter – Fully Safe Mode

Ye sabse safe mode hai.

Isme:

- ✗ Internet nahi
 - ✗ LAN nahi
 - ✓ Sirf Host aur VMs
-

Visual

Host 



VM1  ↔ VM2 

Bas itna hi.

Outside world nahi.

Example

Cybersecurity lab:

Kali Linux (Attacker)

Metasploitable (Victim)

Agar NAT use karoge → victim internet pe expose ho sakta hai.

Agar Bridged use karoge → real network pe expose ho sakta hai.

👉 Isliye Host-Only best hai.

✅ Host-Only Me Allowed

- ✅ Host → VM
 - ✅ VM → Host
 - ✅ VM ↔ VM
 - ❌ Internet
 - ❌ LAN
-

🔥 Ab Confusion Clear Karte Hain

? NAT aur NAT Network me difference?

NAT	NAT Network
Single VM	Multiple VM
No VM-to-VM	VM-to-VM allowed
Internet yes	Internet yes

? Bridged aur NAT me difference?

NAT	Bridged
Hidden	Visible
Private	Real LAN member
Safer	Exposed

? Host-Only aur NAT me difference?

NAT	Host-Only
Internet yes	No internet
VM isolated	VM connected to Host

🎯 Simple Decision Formula

Agar tum soch rahe ho:

👉 "Mujhe sirf internet chahiye"
→ NAT

👉 "Mujhe 2 VMs ko connect karna hai"

→ NAT Network

👉 "Mujhe VM ko real network me lana hai"
→ Bridged

👉 "Mujhe safe hacking lab banana hai"
→ Host-Only

Real Cyber Lab Setup Example

Beginner

1 Kali Linux
👉 NAT

Intermediate

Kali + Metasploitable
👉 Host-Only

Advanced

Kali + Web Server + DB Server
👉 NAT Network

Sabse Important Concept

VirtualBox networking decide karta hai:

1. VM visible hoga ya hidden
2. Internet milega ya nahi
3. Dusre VMs se baat karega ya nahi
4. Secure hoga ya exposed

Golden Memory Trick

Socho VM ek room hai:

- NAT = Window se internet dekh sakta hai
- NAT Network = Rooms ek dusre se baat kar sakte
- Bridged = Room ka direct gate bahar road pe
- Host-Only = Ghar ke andar locked room